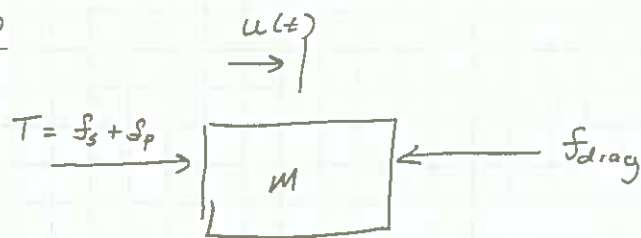


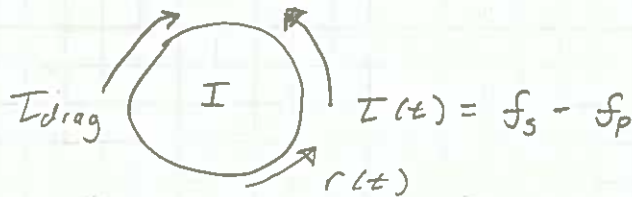
Surge OnlyFBD

$$\rightarrow \Sigma F = m \dot{u} = T - f_{drag}$$

$$f_{drag} = D u$$

$$m \dot{u} + D u = T$$

$$\boxed{\frac{U(s)}{T(s)} = \frac{1}{Ms + D}}$$

Yaw OnlyFBD

$$\rightarrow \Sigma \tau = I \ddot{r} = \tau(t) - \tau_{drag}$$

$$\tau_{drag} = d \dot{r}$$

$$I \ddot{r} + d \dot{r} = \tau(t)$$

$$\boxed{\frac{R(s)}{\tau(s)} = \frac{1}{I s^2 + d s}}$$

$$\text{Let } y(t) = \dot{r}(t)$$

$$I \dot{y} + d y = \tau(t)$$

$$\boxed{\frac{Y(s)}{\tau(s)} = \frac{\dot{R}(s)}{\tau(s)} = \frac{1}{I s + d}}$$